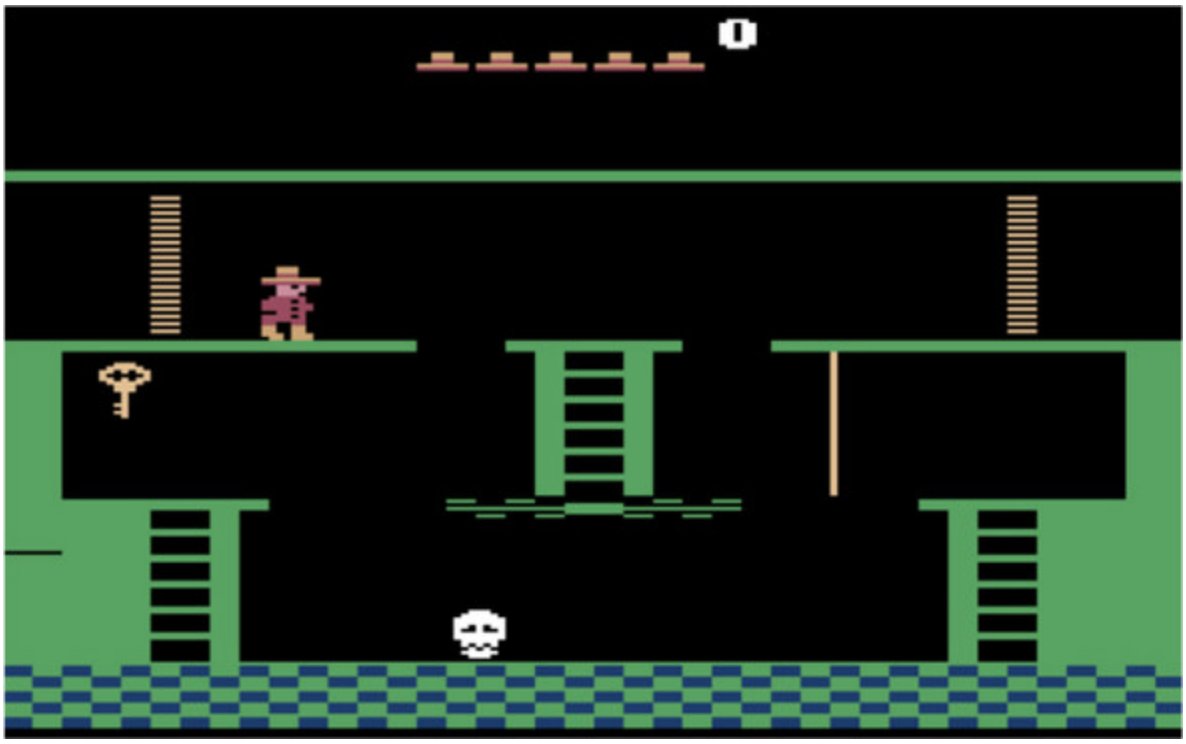


Note

The paper that brought great attention to the field of deep reinforcement learning was “Human-level control through deep reinforcement learning” by Volodymyr Mnih and collaborators at Google DeepMind in 2015. The paper is fairly readable and contains the details you’d need to replicate their results.

What’s the difference between the environments that explains these disparities in performance? The games that DQN was successful at all gave relatively frequent rewards during game play and did not require significant long-term planning. Montezuma’s Revenge, on the other hand, only gives a reward after the player finds a key in the room, which also contains numerous obstacles and enemies. With a vanilla DQN, the agent starts exploring essentially at random. It will take random actions and wait to observe rewards, and those rewards reinforce which actions are best to take given the environment. But in the case of Montezuma’s Revenge, it is extremely unlikely that the agent will find the key and get a reward with